

Adam Optimizer Cheat Sheet

- Adam is best used for large non stationary datasets (Properties change over time)
- Adam already includes exponential decay maintenance for internal momentum
- This exponential decay is not used for the learning rate, so an external exponential decay hyperparameter is often used to suit that purpose
- Anything with a hat (e.g, $\hat{t}$ or $\hat{m}$) is bias corrected (formula 4)

Main Update Formula (1)

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

- First and second movements first bias corrected (see formula 4)
- **After** bias correction these are plugged into the main update formula (formula 1)
- θ
 - Model parameters (weights and biases)
- α
 - Learning rate
- m
 - Bias corrected first moment (has a little hat)
- v
 - Bias corrected second movement (has a little hat)
- ϵ
 - Small value to prevent division by zero (see below)
 - Typical values:
 - In the original Adam paper: $\epsilon = 10^{-8}$
 - Other implementations (e.g., TensorFlow, PyTorch) often use 10^{-8} or 10^{-7}
 - Some optimizers (like AdamW, RMSProp variants) may use values between 10^{-4} and 10^{-8} , depending on the problem.

First Moment Estimate (2)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

- $g_t = \nabla_{\theta} L(\theta_t)$
 - The gradient

- In this context L is the loss function
- and theta (θ) are the vectorized parameters of model weights and biases
- β_1
 - Decay rate for the mean (Typically 0.9), averages direction of gradients
- M
 - Mean (moment)

Second Moment (3)

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

- Where v is the second moment
- β_2
 - Decay rate for the variance (typically 0.999), normalizes learning rates

Bias Correction (4)

$$\widehat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\widehat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

These values are then plugged into formula 1